FINITENESS OF THE FIBRE PRODUCT MODULE OVER THE FIBRE PRODUCT RING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Surjectivity of the map from the fibre product ring onto the fibre product of the two rings. Lemma \ref{lemma-module-over-fibre-product-bis}, cited at proof lines 1048-1049. Makes the constructed module a module over the quotient ring.
- Finiteness of a fibre product of modules over a fibre product of rings. Lemma \ref{lemma-finite-module-over-fibre-product}, cited at proof lines 1051-1054. Gives finiteness over the quotient ring.

Source reference. Original lemma 08IH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10815–10815. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 08KM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10890–10890. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 08IG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10814–10814. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0D2H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16382–16382. Navigation pointer to a statement the

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Lemma 0.2. *Let $A, A', B, B', C, C', D, D', I, M', M, N, \varphi$ be as in Lemma 08KM. If N finite over D and M' finite over C', then $N' = N \times_{\varphi,M} M'$ is finite over D'.*

Proof. Recall that $D' \to D \times_C C'$ is surjective by Lemma 08IG. Observe that $N' = N \times_{\varphi,M} M'$ is a module over $D \times_C C'$. We can apply Lemma 0D2H to the data $C, C', D, D', IC', M', M, N, \varphi$ to see that $N' = N \times_{\varphi,M} M'$ is finite over $D \times_C C'$. Thus it is finite over D'. □